



Lesson 2.2: Computer Organization

https://codehs.com/lms/assignment/93877817/lesson_plan

Description	How are computers organized? What are the main components of a computer? In this lesson, we will explore how different organizational structures of computers interact with each other to make computers functional.		
Objective	Students will be able to: • Understand the main parts of a computer • Differentiate the difference between hardware and software • Identify input and output devices • Learn different types of networks		
Activities	2.2.1 Video: Computer Organization 2.2.2 Check for Understanding: Video Quiz 2.2.3 Connection: Draw a Computer 2.2.4 Check for Understanding: What Kind of Device?		
Prior Knowledge	No prior knowledge needed.		
Planning Notes	• Review the slides and the exercises in the lesson. • This is a shorter lesson with time available for review, enrichment or to move ahead to the next lesson. This would be an opportunity to observe and gauge how each student is progressing with the material. • You may want to review the Technical Reading Tips and Technical Writing Tips handouts as students complete this lesson and those thereafter. • There is an additional handout activity about the ergonomics of a computer device setup, which can be used in classroom or as homework.		
Standards Addressed	CSTA 3A Standards <table><tr><th>Name</th><th>Description</th></tr></table>	Name	Description
Name	Description		

3A-CS-02

Compare levels of abstraction and interactions between application software, system software, and hardware layers.

PRAXIS 5652 Standards

Name	Description
IV.B.3.b	identify the types of data generated by surveys and sensors
V.A.2.c	describe how sensors are used in embedded systems
V.A.3	Know the capabilities, features, and uses of different types of computing systems (e.g., desktop, mobile, cluster)
V.A.3.a	identify capabilities, features, and uses for each type of computer system
V.A.3.b	identify criteria to evaluate and compare computing systems
V.A.4	Be familiar with computers as layers of abstraction from hardware (e.g., logic gates, chips) to software (e.g., system software, applications)
V.A.7.c	identify and compare network types
V.B.1.a	identify network hardware devices and their functions

Teaching and Learning Strategies

Lesson Opener:

- Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins]

Activities:

- Watch the lesson video and take the corresponding quiz. This quiz is a quick check for understanding. [10 mins]
- Complete the *Draw a Computer* connection activity. [10 mins]

- Important note: You will not be able to access students' drawings. If you want to see student drawings, you can create a free-response activity that has students export their drawing, upload it to CodeHS, and then copy and paste the link into the activity. You can learn how to upload images [here](#).
- This is a fun activity to get students thinking about the components of a computer.
- Have students color input devices one color and output devices another color.
- Complete the *What Kind of Device?* quiz. [5 mins]

Lesson Closer:

- Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]

Discussion Questions

Beginning of class:

- What are the main parts of a computer?
 - *Monitor, keyboard, CPU, mouse, etc.*
- What is the difference between hardware and software? What do you know about each?
 - *Hardware is a physical component of a computer whereas software is a computer program.*
- Name a software program.
 - *Microsoft Word, Windows Vista, etc.*

End of class:

- Give an example of an input device.
 - *Microphone, webcam, keyboard, mouse, etc.*
- Give an example of an output device.
 - *Speaker, monitor, printer, etc.*
- What is the difference between a LAN and WAN? How do these two networks interact?
 - A LAN is a local area network and connects computers that are close to each other. They can usually be connected with only one router. A WAN is a wide area network and connects computers that are far from each other. Multiple routers are needed in a WAN.*

Resources/Handouts

[Technical Reading Tips](#)

[Technical Writing Tips](#)

[Computer Ergonomics \(Student\)](#)

[Computer Ergonomics \(Teacher\)](#)

Vocabulary

Term	Definition
Computer	A person or device that makes calculations, stores data, and executes instructions according to a program.
Hardware	The physical components of a computer
Software	Programs that can be run on a computer

Modification: Advanced	Modification: Students Needing Additional Support	Modification: English Language Learners
	• Print out video slides for students to reference	• Print out video slides for students to reference